Inference Investigation (HetGP)

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1 Test Problem

The VHGP is fitted to a heteroscedastic test function shown in Figure 1. The dataset is composed of $k = 25$ unique locations, with $n = 2$ replicates taken at each point.

Table 1: Summary of experiment's parameter values.

Parameter	Value
k	25
n	2
seed	12345

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Text(0.5, 0.98, 'Test Problem')
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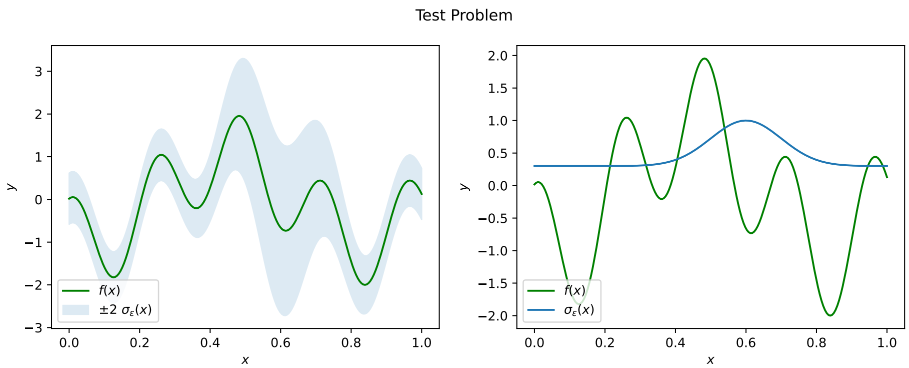
(a) The target function of which noisy data we condition our HetGP on.



(b)

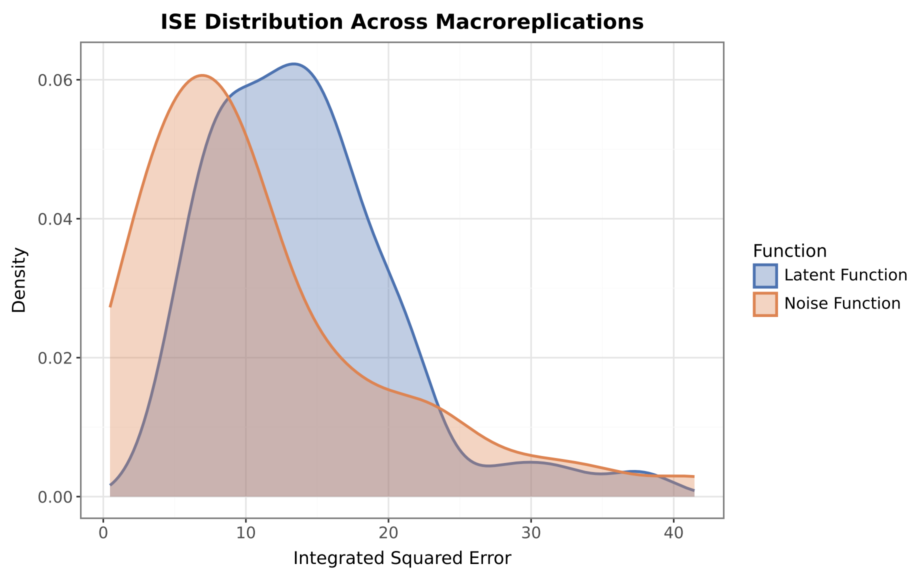
Figure 1

2 Results

Included in this section is the table Table 2; summarising the model likelihood, the ideal likelihood - the likelihood if the truth was our model -, and the integrated square errors for both the noise and latent predictions.

Table 2: Summary of log-likelihoods and integrated square errors

	ll	ideal_ll	ISE_f	ISE_eps
count	50.000000	50.000000	50.000000	50.000000
mean	-13.111374	-27.359180	14.077221	11.252206
std	8.368128	5.314669	6.685112	9.044232
min	-28.608259	-40.149440	5.233377	0.472483
25%	-19.715270	-29.883826	8.930544	5.288328
50%	-13.495062	-26.896718	12.961042	8.620324
75%	-8.335875	-24.419173	16.547824	15.155256
max	7.760140	-16.098354	37.591038	41.457397

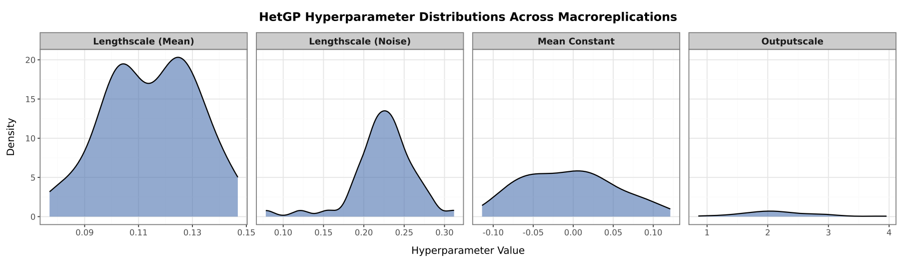


(a)

Figure 2: The Integrated Square Error (ISE) for both noise and latent function, after $M = 50$ macroreplications.

Table 3: Summary stats for HetGP hyperparameters

	l_f	l_g	tau	mu
count	50.000000	50.000000	50.000000	50.000000
mean	0.114749	0.224271	2.184858	-0.005814
std	0.016902	0.038768	0.634623	0.057044
min	0.076967	0.078441	0.855841	-0.113909
25%	0.102631	0.211037	1.781183	-0.049711
50%	0.117345	0.229011	2.128062	-0.010171
75%	0.126630	0.244657	2.575646	0.030182
max	0.146834	0.311859	3.958267	0.121303



(a)

Figure 3: A plot of hyperparameter densities after 50 macroreplications.

3 Appendix

Table 4: Contains the raw hyperparameter data from the experiments

l_f	l_g	tau	mu
0.126474	0.265536	2.239151	-0.090727
0.092234	0.213777	1.483496	-0.089738
0.126682	0.253910	2.418303	0.025816
0.130673	0.229213	2.813188	-0.036773
0.103897	0.121314	2.165163	-0.030033
0.108411	0.194731	1.771776	-0.049959
0.126104	0.231362	2.147608	-0.048840
0.132399	0.240336	2.726211	-0.000427
0.097685	0.190400	2.011147	-0.012188
0.094628	0.189604	1.739429	0.019856
0.122186	0.248464	2.707553	0.011533
0.121669	0.212658	2.062326	0.064079
0.142502	0.155060	3.795379	0.054111
0.129485	0.282202	2.156629	-0.081460
0.101930	0.219644	1.346180	0.024304
0.146834	0.261813	3.958267	-0.080257
0.086670	0.200243	0.855841	-0.048966
0.118506	0.222351	2.292595	-0.012314
0.100427	0.181984	2.393512	0.005131
0.103712	0.230024	1.941169	0.079617
0.101161	0.190342	2.223329	-0.068070
0.102271	0.208427	1.428585	0.014192
0.076967	0.078441	1.702693	-0.045582
0.105580	0.224042	2.049795	-0.084110
0.110313	0.232917	1.609499	0.043460
0.123887	0.215007	2.828066	-0.017062
0.136597	0.264012	2.087208	0.031455
0.111987	0.238115	2.323813	0.023901
0.111202	0.244755	1.976601	-0.035729
0.079741	0.200275	0.969627	-0.008518
0.144810	0.214643	2.962082	0.031943
0.127695	0.231810	2.621730	0.045640
0.122294	0.228810	1.851348	0.095352
0.105548	0.202266	2.024346	0.096599
0.131687	0.221106	2.108515	-0.039670
0.105097	0.236510	1.746085	-0.054956
0.120097	0.213772	2.866580	-0.052744
0.104376	0.276178	1.528071	0.003817
0.087832	0.311859	1.756290	0.053163
0.116185	0.238003	1.876525	-0.011823
0.098395	0.244361	1.809402	-0.013068
0.100695	0.229299	1.223067	0.026363
0.138491	0.210496	3.044771	0.007024
0.120756	0.279700	2.149543	0.077850
0.132493	0.215857	2.829530	0.121303
0.135159	0.262749	2.904620	0.084254
0.123175	0.213939	2.212648	-0.113909
0.104476	0.238945	1.821106	-0.070994
0.125653	0.248514	3.245098	-0.073018
0.119717	0.253786	2.437393	-0.060519

Table 5: Contains the raw metric data from the experiments

m	ll	ideal_ll	ISE_f	ISE_eps
0	-10.639401	-24.551760	11.265863	0.925000
1	-8.033678	-26.162416	8.158135	3.730431
2	-21.517428	-25.179024	9.345174	2.717883
3	-4.351256	-16.098354	15.132927	10.727194
4	-6.904464	-24.622196	8.423431	22.473153
5	-11.455127	-26.086752	13.877617	30.988214
6	-24.493741	-29.934996	8.402217	11.845939
7	-10.012839	-20.630045	12.927650	3.745841
8	-8.808944	-27.894533	12.686035	12.398124
9	-9.148846	-26.816933	16.016164	24.319163
10	-14.335790	-26.368792	10.657472	8.030594
11	-26.028351	-26.799770	6.930100	7.811915
12	-20.859581	-32.781734	19.309035	5.526787
13	-19.417852	-26.784147	12.078821	8.411576
14	-4.277530	-27.018740	28.069854	9.534607
15	-14.588774	-21.719532	5.377360	10.580640
16	-15.559242	-29.730314	17.175547	5.592930
17	3.093726	-22.813566	12.812485	6.747338
18	3.946397	-16.213123	7.246879	15.507192
19	-16.081388	-31.325270	13.365056	9.909288
20	3.341003	-25.278193	21.642202	16.527875
21	-15.253053	-30.328964	9.506686	9.139988
22	-19.865395	-37.870003	21.957969	22.581035
23	-12.654334	-23.193047	7.840118	7.809462
24	-9.918554	-28.331950	21.434956	18.831442
25	-22.637780	-32.186245	12.084087	8.583664
26	-15.427347	-22.175262	12.994434	3.586781
27	-8.178185	-21.665836	5.813095	9.908994
28	-12.328996	-25.656046	31.869977	5.247703
29	7.760140	-26.976503	16.395533	41.457397
30	-28.608259	-28.991657	18.174512	8.656984
31	-11.134919	-21.199543	16.598588	11.777705
32	-22.354457	-27.932741	8.147451	4.062724
33	-3.763512	-28.656940	12.283041	14.099445
34	-15.333731	-30.359413	15.898936	5.906263
35	-19.480953	-40.149440	8.923463	27.503920
36	-12.565086	-20.789412	7.222705	6.010521
37	-22.553184	-30.737316	19.361973	0.472483
38	-22.504820	-39.899376	11.272436	2.031874
39	-22.136313	-33.772340	15.811534	4.528260
40	-2.613886	-26.995710	14.236457	22.074746
41	-6.197974	-29.263310	20.710356	34.781645
42	-9.143585	-21.749147	5.233377	17.873526
43	-19.785525	-28.727955	8.951785	5.410203
44	-24.321091	-26.111784	5.496015	10.515554
45	-17.288639	-28.906195	20.338589	1.489652
46	-17.657044	-37.452810	37.591038	1.725864
47	-4.417654	-24.374977	15.976107	4.784666
48	-9.566941	-22.387623	14.752485	15.826771
49	-19.504506	-36.307262	16.083336	7.879335